

dynLattice: A finite element environment for dynamic simulation of beam networks and lattice metamaterials

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Summary

Beams and beam structures undergoing fast deformation often experience large strains and inelastic material behaviour. Understanding this behaviour enables engineers and designers to design beam networks resulting in so-called mechanical metamaterials. Metamaterials offer unique properties not found in naturally occurring materials. Harnessing these properties presents new possibilities for a wide range of fields (Davies et al., 2025; Jiao et al., 2023). In order to enable engineers to design mechanical metamaterials, the relation between the geometry of the beam network and the resulting properties needs to be understood. For large deformations at high rates, this usually requires destructive testing, limiting the speed of developments. dynLattice presents an efficient open-source numerical tool for investigating the dynamic behaviour of beam networks and lattice metamaterials. Combining geometrically exact beam elements, inelastic material behaviour, contact, and explicit adaptive time integration, it can accurately simulate nonlinear deformations at high rates and reduce the reliance on physical experiments during the design process.

Background

The following is a short description of the beam kinematics and cross-sectional kinetics implemented in this software, with a more detailed description given in (Gärtner, 2025).

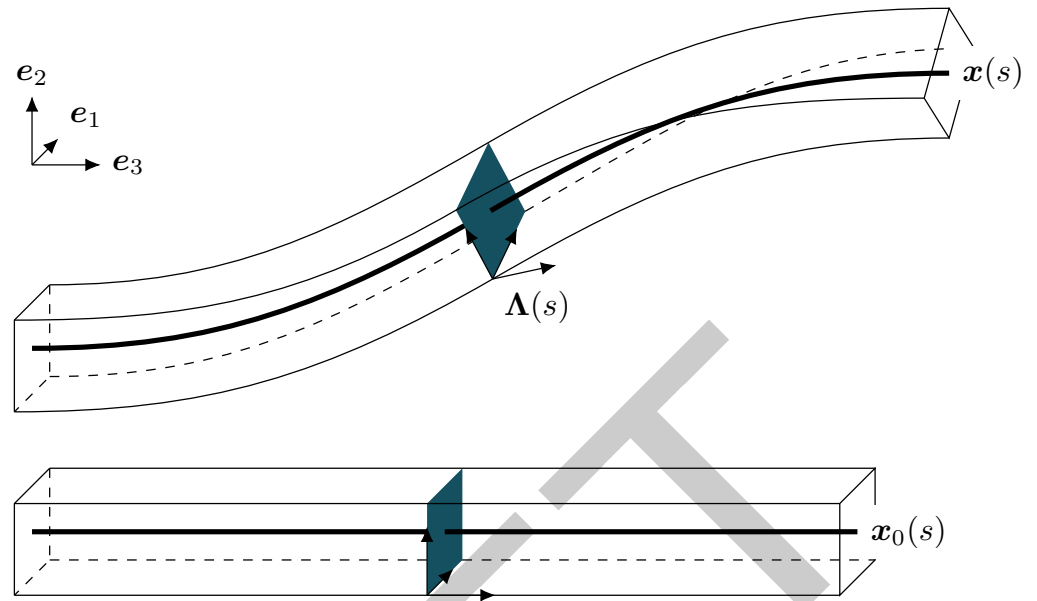


Figure 1: A beam undergoing large deformation.

In Figure 1, a beam undergoing large deformation is shown. A beam is described with the centerline $\mathbf{x}(s)$, where $s \in [s_0, s_1]$ is the measure along the length of the beam, with two orthonormal directors attached to it $\mathbf{d}_1(s), \mathbf{d}_2(s)$. Together with a third orthonormal director $\mathbf{d}_3(s) = \mathbf{d}_1(s) \times \mathbf{d}_2(s)$, these are the column vectors of the rotation matrix between the global reference frame and the local coordinate frame of the beam:

$$\Lambda(s) = \mathbf{d}_i(s) \otimes \mathbf{e}_i = [\mathbf{d}_1(s) \quad \mathbf{d}_2(s) \quad \mathbf{d}_3(s)].$$

From these measures, strain prescriptors, describing stretching, shearing, bending, and twisting of the beam can be derived. These strain prescriptors are then used together with the material model of the beam, be it elastic (cf. (Eugster, 2015; Simo & Vu-Quoc, 1986)) or elasto-plastic (cf. (Gärtner, van den Boom, et al., 2025a; Herrnböck et al., 2022; Smriti et al., 2021)), to assemble the global force vector and—in the static, implicit case—the tangent stiffness matrix using standard finite element procedures.

Statement of need

Research on the dynamic behaviour of mechanical metamaterials, especially beam lattices, relies on numerical tools that can capture large deformations, inelastic behaviour, and contact while remaining efficient and flexible enough for parameter studies (Bonfanti et al., 2024). General-purpose finite element frameworks, e.g. (Dadvand et al., 2010; Ferrándiz et al., 2025; Richart et al., 2024), provide broad capabilities but often lack a complete combination of non-linear Timoshenko-Ehrenfest beam formulations, robust explicit time integration, and inelastic constitutive behaviour. Here, a specialized lattice framework, e.g. (McDonnell & Ning, 2022), typically delivers high-fidelity beam kinematics but only focuses on elastic deformations or omits contact between the constituting beams of the lattice. Resulting from this, many researchers rely on their own implementation of Simo-Reissner beams, which are usually not available to the community and only consider a subset of the requirements needed for a general toolkit. An open and extensible environment that unifies geometrically exact beams, explicit dynamics, inelasticity, and contact is therefore central to advancing simulation-driven design of lattice metamaterials for high-rate and large deformation applications. Here, dynLattice takes a first step by building upon the JIVE framework (Dynaflow Research Group, 2021;

50 [Nguyen-Thanh et al., 2020](#)), keeping the extensible object-oriented design philosophy, and
51 providing extensive documentation in the style of the underlying framework.

52 State of the field

53 There is a wide plethora of existing finite element toolkits available in the market, such as
54 MOOSE ([Lindsay et al., 2022](#)), FEniCS ([Baratta et al., 2023](#)), or deal.II ([Arndt et al., 2023](#)),
55 to name several prominent ones. The usage of these libraries for explicit dynamics with inelastic
56 beams however, would require substantial alteration of their code, as for example in FEniCS
57 explicit dynamics is only available using lumped mass matrices, which would be non-physical for
58 the rotational inertia contributions of the beams, in deal.II neither beam elements nor explicit
59 solves are available and in MOOSE only a limited Timoshenko-Ehrenfest beam is implemented
60 with only elastic behaviour. Other finite element toolkits having built in beam-elements and
61 explicit dynamics include Kratos ([Dadvand et al., 2010](#); [Ferrández et al., 2025](#)) and Akantu
62 ([Richart et al., 2024](#)). For Kratos similar problems as for MOOSE can be seen in only a
63 limited set of beam elements being available and none supporting inelastic behaviour in the
64 cross-sectional strain measures. Akantu, focusing on fracture mechanics and providing a solid
65 representation of contact, on the other hand only provides Euler-Bernoulli beam elements,
66 neglecting all shear deformation within the elements. Next to general finite element toolkits,
67 there are specialized beam toolkits, most prominently for shear-deformable beams, GXBeam
68 ([McDonnell & Ning, 2022](#)). Here, the same beam theory employed in this contribution is used,
69 however due to its focus on composite beams, it is limited to (anisotropic) elastic material
70 behaviour and does not model contact.

71 Software design

72 In order to design metamaterials undergoing large deformations at high rates, dynLattice
73 provides both researchers and engineers in the field a toolkit for numerical experiments.
74 dynLattice is built on top of the openly available C++ JIVE finite element toolkit ([Dynaflow
75 Research Group, 2021](#); [Nguyen-Thanh et al., 2020](#)) and follows in its design the object-
76 oriented framework of JIVE to describe the partial differential equations and corresponding
77 solution procedures. The toolkit allows for the implementation of original methods through the
78 development of custom *modules*, for parts of the program flow in the simulation framework,
79 and of custom *models*, for anything related to the numerical representation of the system
80 of partial differential equations. In the present addition to the JIVE framework, the *models*
81 further contain a separate *materials* class, representing the constitutive models considered in
82 the problem, separating geometric from material considerations. This separation allows both
83 the easy and user-friendly extension of the frameworks by simple inheritance of the provided
84 classes as well as usability by structuring the inputs needed for the program flow and the
85 problem description. dynLattice additionally uses GMSH ([Geuzaine & Remacle, 2009](#)) to mesh
86 the imported geometries efficiently.

87 Research impact statement

88 dynLattice has enabled a series of research efforts investigating lattice metamaterials ([Gärtner
89 et al., 2024, 2025b](#); [Gärtner, 2025](#)). In addition to investigating lattice materials, it has been
90 used to study the behaviour of beams as a multi-scale problem ([Gärtner, van den Boom, et al.,
91 2025a](#)) and has undergone a thorough comparison with experimental data and results from
92 commercial tools ([Gärtner, Dekker, et al., 2025](#)). Collaborating researchers could easily install
93 and run the software using the provided documentation and apptainer.

94 In addition to external collaboration, dynLattice was also used as a ready-made software
95 package by multiple student theses ([IJendoorn, 2024](#); [Niessen, 2022](#); [Smit, 2024](#)). During

these projects, dynLattice was used on a large cluster ((DHPC), 2024) and on various local machines using the provided apptainer. This demonstrates its availability and usability as stand-alone software without the need for further modifications.

Changes to the source code that enable further research are made possible by the object-oriented philosophy inherited from the JIVE package and the clear structure of the code into *modules*, *models* and *materials*, as described in Software design. The provided documentation, available in the source and on the project website, facilitates quick and straightforward adoption by third parties. The included benchmarks serve as a verification tool for the correctness of the implementation and as a guide for creating and extending simple problems.

AI Usage Disclosure

During the creation of this software, GitHub Copilot (using different versions of Claude Sonnet) was used to enhance the documentation of the source code. In particular, it helped streamline all documentation efforts into Doxygen compatible comments. All generated documentation strings were read and approved by the author.

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